

[Chathurya V R 251801045-2025-AIDS](#) ▾**C2****Started on** Thursday, 17 September 2026, 8:05 AM**State** Finished**Completed on** Thursday, 17 September 2026, 8:18 AM**Time taken** 12 mins 49 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
```

```
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
```

```
    {
```

```
        i++;
```

```
        s += i;
```

```
    }
```

```
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n;
5      // No counter increment needed for scanf per the instructions
6      if (scanf("%d", &n) != 1) {
7          return 1;
8      }
9
10     int count = 0;
11
12     // Initializations (Assignments count as 1 step each)
13     int i = 1; count++;
14     int s = 1; count++;
15
16     while (s <= n) {
17         count++; // Incremented for a successful while-condition check
18         i++;     count++; // Incremented for body statement 1
19         s += i;  count++; // Incremented for body statement 2
20     }
21     count++; // Incremented for the final while-condition check that evaluates to false
22
23     // Output the total step count
24     printf("%d", count);
25
26     return 0;
27 }
28

```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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